Name: ______________________________

Brought to you by the Bootstrap team:

- Emmanuel Schanzer
- Kathi Fisler
- Shriram Krishnamurthi
- Dorai Sitaram
- Joe Politz
- Ben Lerner
- Nancy Pfenning
- Flannery Denny
- Rachel Tabak

Modeling Data

A Quick Review…

When viewing a cloud of points on a scatter plot, sometimes we can see a pattern in the data.

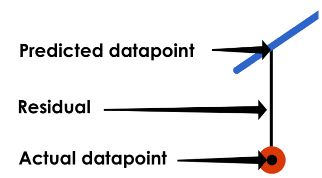
- If the points cluster around a straight line, it might mean there's a ***linear relationship*** between the ***explanatory variable*** (x) and ***response variable*** (y).
- The line can slope up or down, indicating a *positive* relationship (where the two variables increase together) or *negative* relationship (where the response variable decreases as the explanatory variable increases).

These lines (that we see the point clouds clustering around) are known as ***models*** for the data. The straight-line function describing a linear relationship is called a *linear model*.

- With a good model, the point cloud will hug the line tightly. A poor model will have lots of points that stray far from the line.
- Models *summarize* the data. For most datasets that means there will be data points that are not *exactly* on the line! And sometimes the line of best fit won't even pass through a single point in the dataset.
- We can use **linear regression** (`lr-plot` in Pyret) to compute the *best possible linear model* for a dataset, known as the **line of best fit**.

S: Measuring Error in a Model

Differences between the predicted y-value and actual y-value for each x-value are called ***residuals***.
A residual tells us "how wrong" the model was at that particular point.



$$\text{Data} = \text{Model} + \text{Error}$$

We can summarize the error for *all* the points in a dataset using the ***Standard Deviation of the Residuals*** — known as ***S*** — to get a sense of how much to trust the predictions made by a model.

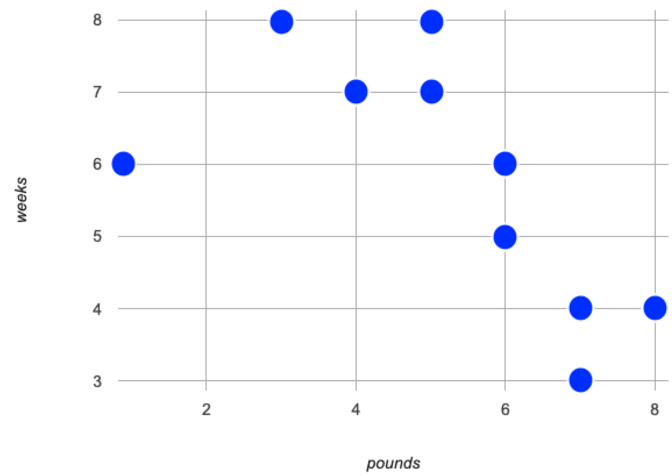
- ***S*** is expressed in terms of *units of the response variable* (y) and tells us how much error we expect in predictions made from the model. *(e.g. up to $8000, 5 years, 11 inches, etc.)*
- The closer the data points are to the model, the smaller the residuals are, and the smaller ***S*** will be.
- If the ***S-value*** for a model is zero, *it fits the data perfectly!*
- When we compare two models for the same dataset, the one with the lower ***S-value*** fits better.
- We have no way of knowing whether or not ***S-values*** represent a small or large amount of error until we consider them in relation to the range of the dataset! *(e.g. errors of $20,000 are huge in the context of average teacher salary, but small in the context of national budgets.)*

How could we Measure Whether a Model is a Good Fit? (Lizards)

Summarize the Relationship You See

Below is a sample of lizards from the Animals Dataset.

name	pounds	weeks
Amy	5	7
Aries	4	7
Boss	6	5
Brittany	3	8
Buck	7	4
Butterscotch	1	6
Chico	8	4
Coconut	6	6
Dodger	5	8
Dylan	7	3



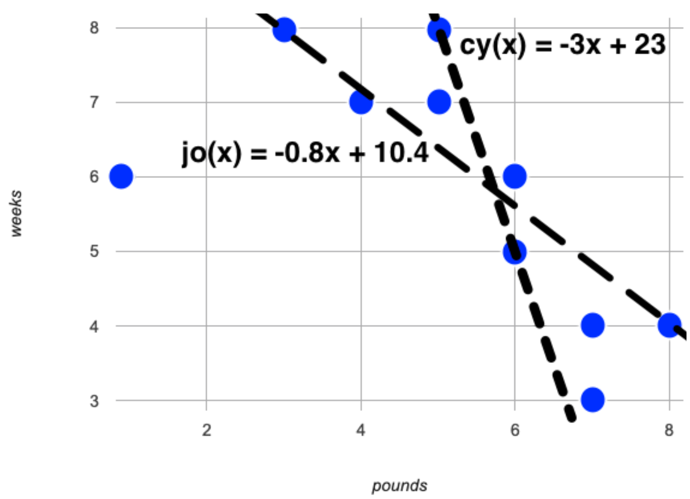
1) Use a straightedge to draw the ***line of best fit*** that best summarizes the relationship you see in the data on the scatter plot.

2) Describe how you decided where to draw the line. ______________________________

____

Comparing Models

3) Cy and Jo drew the two lines below. Do you think `cy(x)` or `jo(x)` is a better model for this data? Why?



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4) What could we measure, to calculate *how much better of a model* it is? ______________________________

____

____

5) *Neither of these models is the best possible model!* What would have to be true of a third model, for us to know that it was a better fit than these two? ______________________________

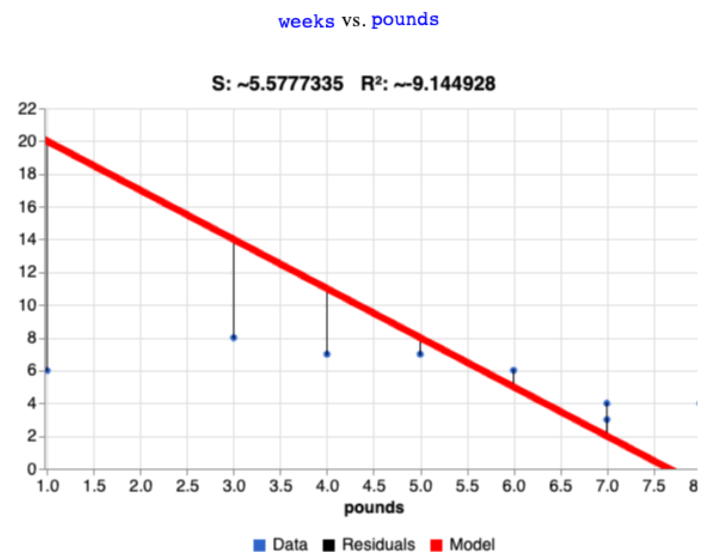
____

____

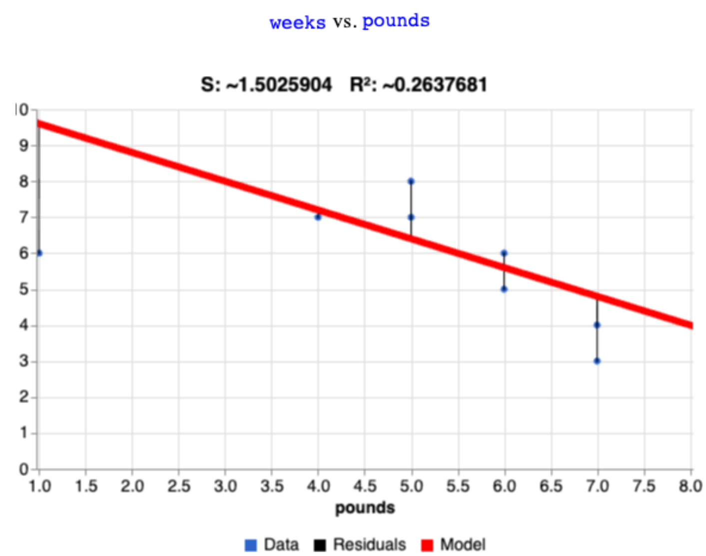
Introducing fit-model

These data visualizations were generated using the [Lizard Sample Starter File](). They can be viewed as interactive charts by uncommenting the final lines in the Definitions Area and clicking "Run".

```
fun cy(x): (-3 * x) + 23 end
fit-model(
 lizard-sample, "name", "pounds", "weeks", cy)
```




```
fun jo(x): (-0.8 * x) + 10.4 end
fit-model(
 lizard-sample, "name", "pounds", "weeks", jo)
```



What do you Notice?	What do you Wonder?

1) How is the `fit-model` plot for cy's model **similar** to the `fit-model` plot for jo's model? ______________________________

____

____

2) How is the `fit-model` plot for cy's model **different** from the `fit-model` plot for jo's model? ______________________________

____

____

3) What do you think the three terms in the legend refer to?

- Model: ______________________________
- Data: ______________________________
- Residuals: ______________________________

Considering *S* in Context

For each model below, decide whether you agree that the model is a good fit. *Then* rank the models from 1 (best fit) to 8 (worst fit).

How good is the model?	Ranking
1) A data scientist is working with data from animals at a shelter. • The range of days to adoption in this dataset are from 0 to 400. • An S value of 300 means predicted adoption times could be off by 300 days. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
2) A student is exploring a dataset on climate change. • The range of Arctic Sea Ice is from 3,920,000 to 7,670,000 square kilometers • An S value of 300 means predicted Arctic Sea Ice coverage could be off by 300 square kilometers. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
3) A data scientist is working with data from US public schools. • The range of graduates per school per year is 2 to 2003. • An S value of 300 means predicted graduate values could be off by 300 students. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
4) A student is exploring a dataset on earthquakes. • The range of earthquake depths in this dataset are from 4200m to 664000m. • An S value of 300 means predicted earthquake depths could be off by 300 meters. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
5) A student is exploring a dataset on arrests in Los Angeles. • The age range in this dataset is from 0 to 92. • An S value of 1 means predicted ages could be off by 1 year. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
6) A data scientist is working with data about snowflakes. • The range of snowflake weights is from 0.001 grams to 0.02 grams. • An S value of 1 means predicted values could be off by 1 gram. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
7) A data scientist is working with data from animals at a shelter. • The range of ages is from 0.5 years to 16 years. • An S value of 1 means predicted ages could be off by 1 year. I ______________________ that this model is a good fit. strongly agree, agree, disagree, strongly disagree	
8) A student is working with a dataset of adult blue whales. • The range of weights is 200,000 to 330,000 pounds. • An S value of 1 means predicted weights could be off by 1 pound. I ______________________ that this model is a good fit strongly agree, agree, disagree, strongly disagree	

Interpreting our Models

Describe the Data Used to Build the Models

1) These models were built using data about ________ (how many?) ____________________ (what?) whose

____________ (x-variable) ranged from ______ (x-min) ____________ (x-variable units) to ______ (x-max) ____________ (x-variable units).

2) Possible limitations of this model include ________________________________ (uneven distribution, high variability, small sample size, etc.)

Cy's Model: $cy(x) = -3x + 23$

3) **This model predicts** that lizards weighing 0 pounds will be adopted in ______ (y-intercept) ____________ (y-units) and that,

for every additional ___pound a lizard weighs___ (x-variable units), ____________ (y-variable) will ____________ (increase/decrease) by ______ (rate of change) ______ (y-units).

4) **The error in the model** is described by an ***S-value*** of about ______ (S) ____________ (y-units). I ____________________ (strongly agree, agree, disagree, strongly disagree)

that this model is a good fit considering that ____________ (y-variable units) in this dataset range from ________ (lowest y-value) to ________ (highest y-value).

Jo's Model: $jo(x) = -0.8x + 10.4$

5) **This model predicts** that lizards weighing 0 pounds will be adopted in ______ (y-intercept) ____________ (y-units) and that,

for every additional ____________ (x-variable units), ____________ (y-variable) will ____________ (increase/decrease) by ______ (rate of change) ______ (y-units).

6) **The error in the model** is described by an ***S-value*** of about ______ (S) ____________ (y-units). I ____________________ (strongly agree, agree, disagree, strongly disagree)

that this model is a good fit considering that ____________ (y-variable units) in this dataset range from ________ (lowest y-value) to ________ (highest y-value).

Comparing Models

7) Is Jo's model better or worse than Cy's model? ________________________________

8) How much ________ (more / less) error do we expect in predictions made with Jo's model than predictions made with Cy's model?

$$\text{Percent Change} = \frac{\text{Difference between the S-values}}{\text{S-value for Cy's model}} \times 100 = \_\_\_$$

We expect predictions made with Jo's model to have ______ percent ________ (more / less) error than predictions made with Cy's model!

My Model

If your teacher had you complete From Lines to Functions, write the function you defined for your model on the line below and then complete the interpretation. If your class did not define models for the lines you drew, you can skip this section.

$\text{my-model}(x) =$ ________________________________

9) **This model predicts** that lizards weighing 0 pounds will be adopted in ______ (y-intercept) ____________ (y-units), and that,

for every additional ____________ (x-variable units), ____________ (y-variable) will ____________ (increase/decrease) by ______ (rate of change) ______ (y-units).

10) **The error in the model** is described by an ***S-value*** of about ______ (S) ____________ (units). I ____________________ (strongly agree, agree, disagree, strongly disagree)

that this model is a good fit considering that ____________ (y-variable units) in this dataset range from ________ (lowest y-value) to ________ (highest y-value).

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